

Executive summary (3 sentences):

We built a conversion engine that generates signal-grounded outbound and measures reply, stalled-th

On ~´†-Bench retail (sealed), the method scores 0.450 (95% CI [0.250, 0.650], n=20) vs Day-1 baselin

Recommendation: run a 30-day Segment 2 pilot with a hard kill-switch on wrong-signal rate, and treat

1) ~´†-Bench baseline performance (sealed, 95% CIs):

- Method pass@1: 0.450 (95% CI [0.250, 0.650], n=20)
- Day-1 baseline pass@1: 0.400 (95% CI [0.200, 0.600], n=20)
- Auto-opt baseline pass@1: 0.400
- Published reference (point): 0.42
- ~´† task duration: p50=27.46s, p95=61.53s (simulator runtime; not email response latency)

2) Cost per qualified lead (CPL):

- Total measured cost (Act V rollup): \$0
- CPL (qualified = booking_created): \$0

3) Stalled-thread rate delta:

- Measured stalled-thread rate: 92.3% (48/52 non-booking)
- Tenacious manual baseline (given): 30.0%â40.0%

Context: this measurement uses booking_created as a proxy gate. A high stalled rate here can reflect

4) Competitive-gap outbound performance (synthetic evaluation):

- Competitive-gap reply rate: 100.0% (25/25)
- Generic reply rate: 19.0% (29/153)
- Reply-rate delta (cg â generic): 81.0%
- Top-quartile real-world signal-grounded range (benchmark): 7.0%â12.0%

Note: 100% reply rates are a synthetic-data artifact in this run; treat as directional signal only until valid

5) Annualized dollar impact (segment-based revenue model):

- Segment 2: assume 2 discovery calls/week â 6.0â15.0 closes/yr
Annual revenue impact: \$1,440,000â\$10,800,000 (range)
- Segment 1, Segment 2: assume 6 discovery calls/week â 18.0â45.0 closes/yr
Annual revenue impact: \$4,320,000â\$32,400,000 (range)
- Segment 1, Segment 2, Segment 3, Segment 4: assume 12 discovery calls/week â 36.0â90.0 closes/yr
Annual revenue impact: \$2,880,000â\$64,800,000 (range)

6) Pilot scope recommendation (30 days):

- Segment: Segment 2
- Lead volume: assume 2 discovery calls/week booked
- Weekly budget cap: \$25 (assumption)
- Success criterion: â\$5 discovery calls booked with qualified Segment 2 prospects within 30 days

1) Wrong-signal outreach (signal_overclaiming):

Why $\sim \uparrow$ misses: $\sim \uparrow$ retail tasks do not exercise public-signal grounding for real companies.

Cost: lower throughput; requires more enrichment checks per prospect.

What: phrasing that reads like low-quality outsourcing spam.

Add: style-guide rubric + red-flag phrase filters + human review for first 50 sends.

Cost: review time; false positives can reduce volume.

What: system can get replies but fail the booking proxy due to tooling mismatch.

Add: align qualification to Tenacious CRM stages; instrument post-reply handoff.

Cost: HubSpot stage mapping + ops time.

What: "top-quartile practice" may be irrelevant or strategically wrong for a prospect.

Add: ask a calibration question before prescribing; allow a deliberate choice escape hatch.

Cost: longer emails; slightly lower reply rates expected.

False negative: quietly sophisticated but silent company â scorers underestimate readiness â overly

False positive: loud but shallow company → scorer overestimates readiness → over-technical pitch →

Risk 1: deliberate strategy (e.g., regulated workflow where "move fast" practices are harmful) → wrong

Risk 2: sub-niche irrelevance (capability gap is not binding constraint) $\hat{\rightarrow}$ prospect ignores message and

Assume 1,000 emails sent; if 5% contain a wrong signal, that's 50 wrong-signal emails.

If each wrong-signal email has an expected reputation cost of \$10k in pipeline suppression, expected

Compare to expected reply-rate lift in the 7â 12% range (vs 1â 3% baseline).

Probe: signal_overclaiming (P-005): assertive Segment 1 opener regardless of confidence (measured

Impact: undermines the core "grounded research" value prop; can revert reply rates to baseline and

Metric: wrong-signal rate on manually audited sample of first 100 sends.

Threshold: pause if wrong-signal rate $\hat{\geq}$ 5% or any single high-severity false claim triggers an executive

Rollback: revert to generic pitch template + require human approval for signal-grounded claims until fix